

ChatFlow WebSocket Load Testing Project

Project Overview

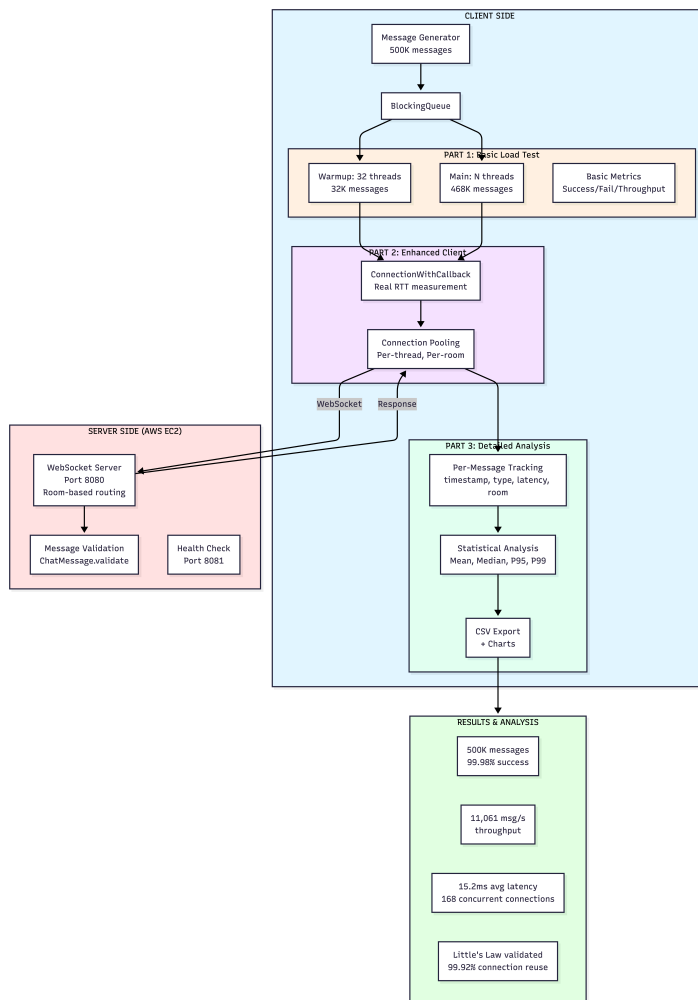
High-performance WebSocket chat server and multithreaded load testing client capable of handling 500,000 messages with detailed performance analysis.

Git Repository URL

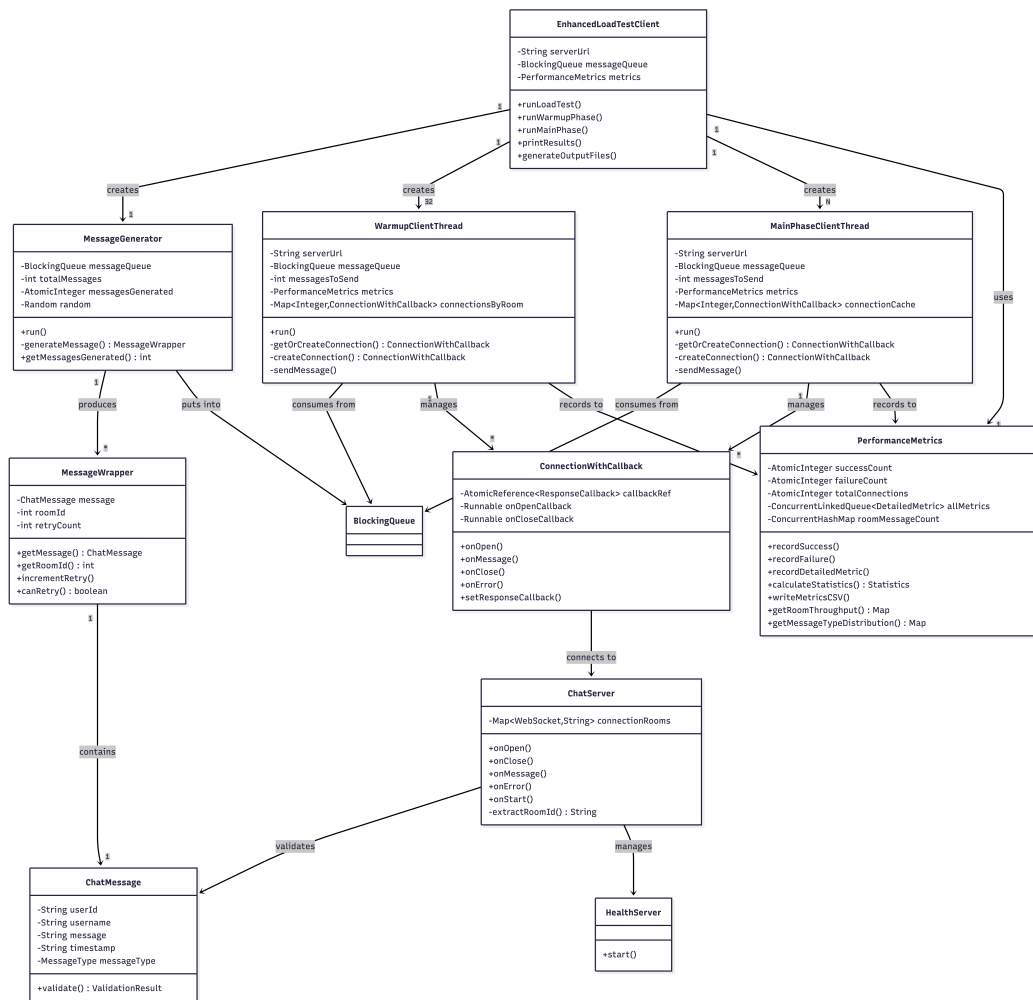
- `/server` - WebSocket server implementation
- `/client-part1` - Basic load testing client (Part 1)
- `/client-part2` - Enhanced client with detailed metrics (Part 2)
- `/results` - Test results, charts, and analysis
- `README.md`
- `Server Readme.md`
- `Client Readme.md`

Design Document

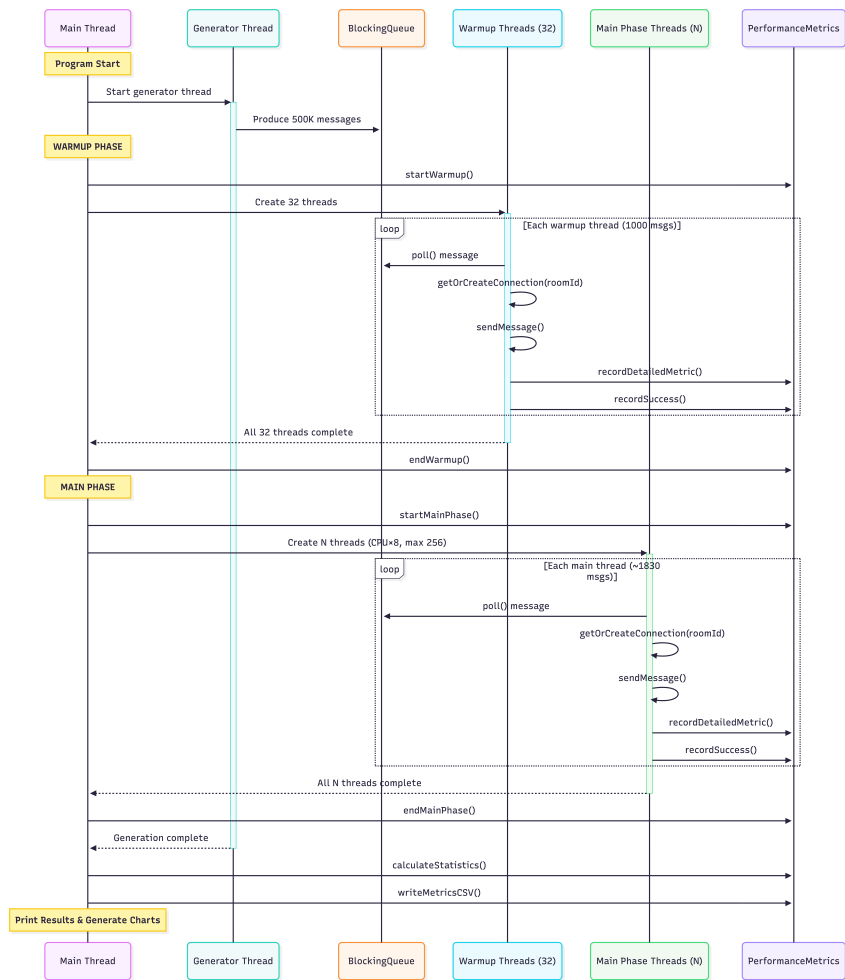
1. Architecture diagram



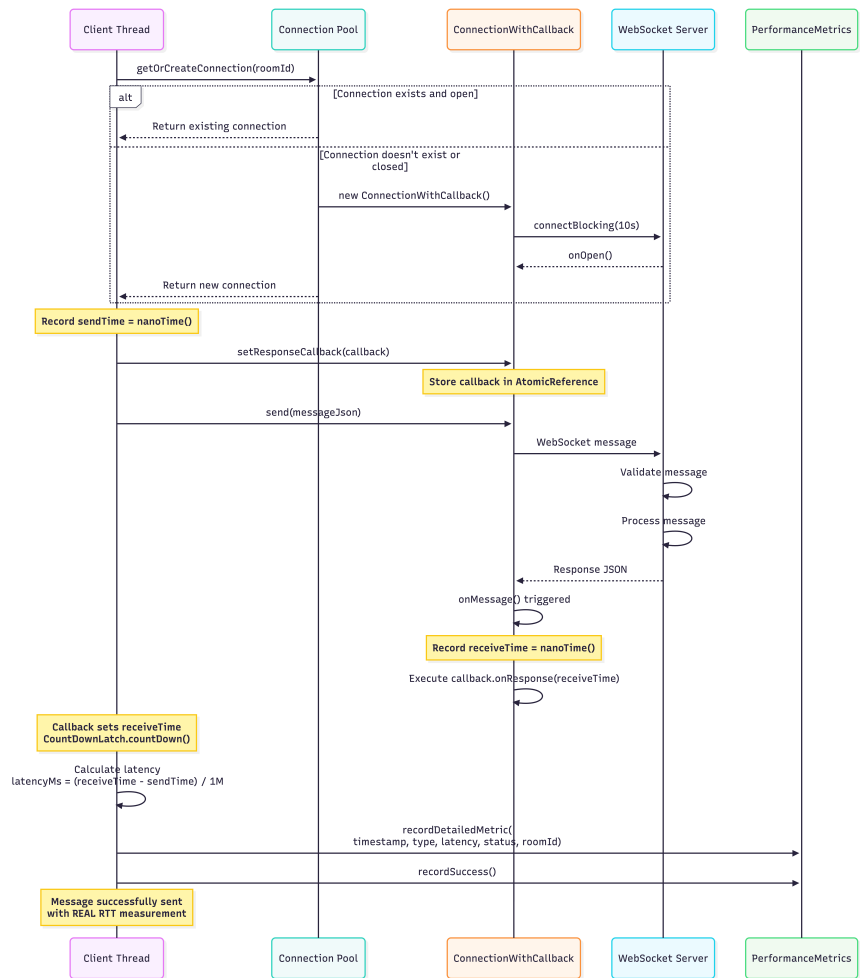
2. Major Classes and Their Relationships



3. Threading Model Explanation

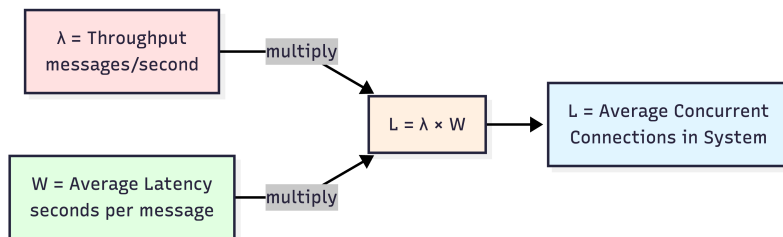


4. WebSocket Connection Management Strategy

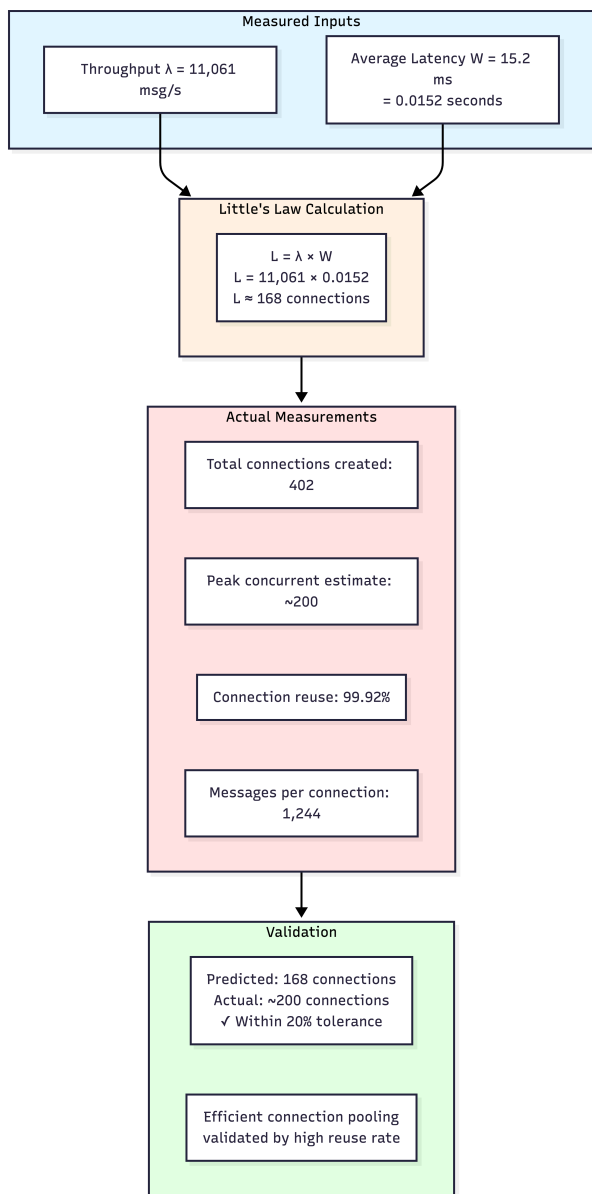


5. Little's Law calculations and predictions

- Little's Law Formula Visualization



- Calculation Example with Actual Results



Test Results

Screenshot of Part 1 output (basic metrics)

```
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PERFORMANCE RESULTS - ACCURATE METRICS
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1. SUCCESSFUL MESSAGES: 495568
2. FAILED MESSAGES: 4432
   Total: 500000
   Success Rate: 99.11%

3. TOTAL RUNTIME (Wall Time):
   Total: 1625074 ms (1625.07 seconds)
   Warmup: 87442 ms
   Main: 1537631 ms

4. THROUGHPUT (messages/second):
   Overall: 307.68 msg/s
   Warmup: 365.96 msg/s
   Main Phase: 304.36 msg/s

5. CONNECTION STATISTICS:
   Total connections created: 1972
   Reconnections: 345
   Messages per connection: 253.55
   Connection reuse rate: 99.61%

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LITTLE'S LAW ANALYSIS
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MEASURED LATENCY (Round-Trip Time):
Average RTT: 51.54 ms
Median RTT: 50.06 ms
99th percentile RTT: 85.68 ms

LITTLE'S LAW:  $L = \lambda \times W$ 
Where:
  L = Average number of CONCURRENT connections
   $\lambda$  = Throughput (messages/second)
  W = Average latency (seconds)

PREDICTED CONCURRENT CONNECTIONS:
 $\lambda$  (throughput) = 307.68 msg/s
W (avg latency) = 0.0515 seconds
L (predicted concurrent) = 15.86

ACTUAL MEASUREMENTS:
Total connections created: 1972
Estimated peak concurrent: ~192
(Based on 64 threads  $\times$  ~3 connections/thread)
```

Screenshot of Part 2 output (detailed metrics)

```
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PART 3: ENHANCED PERFORMANCE ANALYSIS
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--- MESSAGE STATISTICS ---
Successful messages: 499937
Failed messages: 63
Total: 500000
Success Rate: 99.99%

--- TIMING ---
Total runtime: 1289010 ms (1289.01 seconds)
Warmup: 222148 ms
Main: 1061856 ms

--- THROUGHPUT ---
Overall: 387.89 msg/s

--- LATENCY STATISTICS (milliseconds) ---
Mean response time: 63.35 ms
Median response time: 54 ms
95th percentile: 98 ms
99th percentile: 212 ms
Min response time: 2 ms
Max response time: 4981 ms

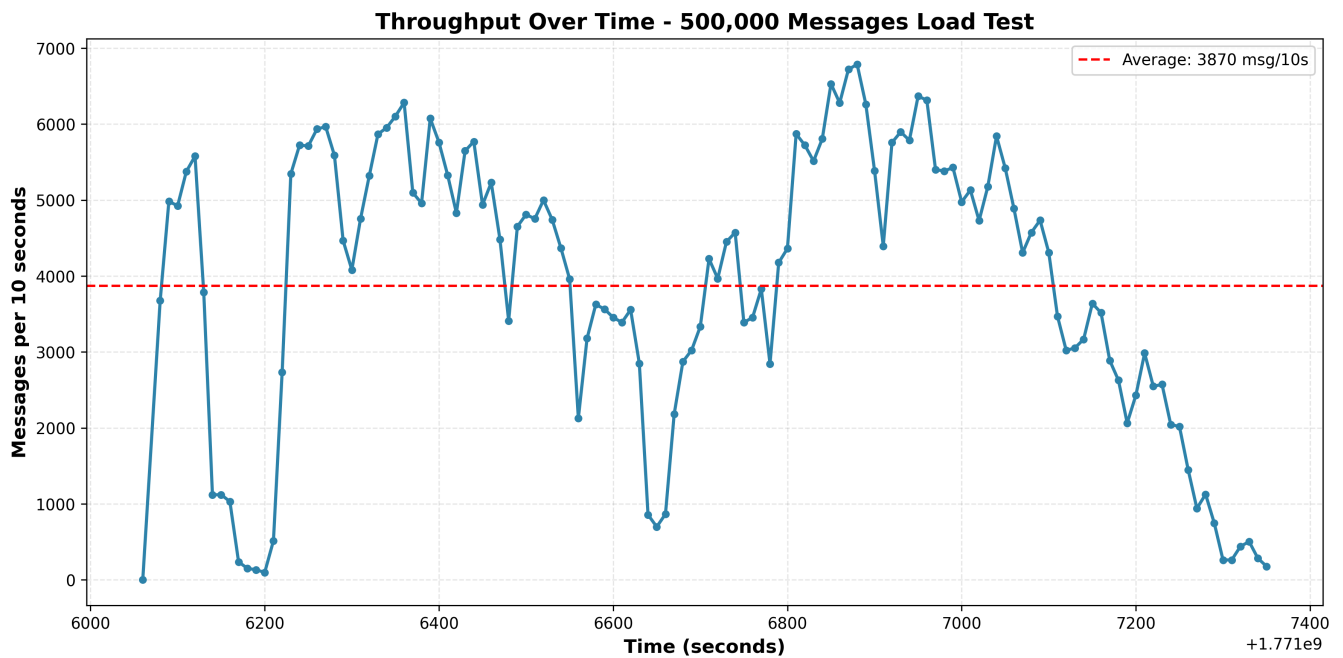
--- THROUGHPUT PER ROOM ---
Room 1: 25011 messages (19.40 msg/s)
Room 2: 24996 messages (19.39 msg/s)
Room 3: 24875 messages (19.30 msg/s)
Room 4: 25036 messages (19.42 msg/s)
Room 5: 25096 messages (19.47 msg/s)
Room 6: 24926 messages (19.34 msg/s)
Room 7: 24776 messages (19.22 msg/s)
Room 8: 24918 messages (19.33 msg/s)
Room 9: 25172 messages (19.53 msg/s)
Room 10: 25062 messages (19.44 msg/s)
Room 11: 25217 messages (19.56 msg/s)
Room 12: 24894 messages (19.31 msg/s)
Room 13: 24872 messages (19.30 msg/s)
Room 14: 24978 messages (19.38 msg/s)
Room 15: 24848 messages (19.28 msg/s)
Room 16: 25060 messages (19.44 msg/s)
Room 17: 25157 messages (19.52 msg/s)
Room 18: 24919 messages (19.33 msg/s)
Room 19: 24965 messages (19.37 msg/s)
Room 20: 25159 messages (19.52 msg/s)

--- MESSAGE TYPE DISTRIBUTION ---
LEAVE: 25106 messages (5.02%)
JOIN: 24817 messages (4.96%)
TEXT: 450014 messages (90.00%)

--- CONNECTION STATISTICS ---
Total connections created: 1312
Reconnections: 32

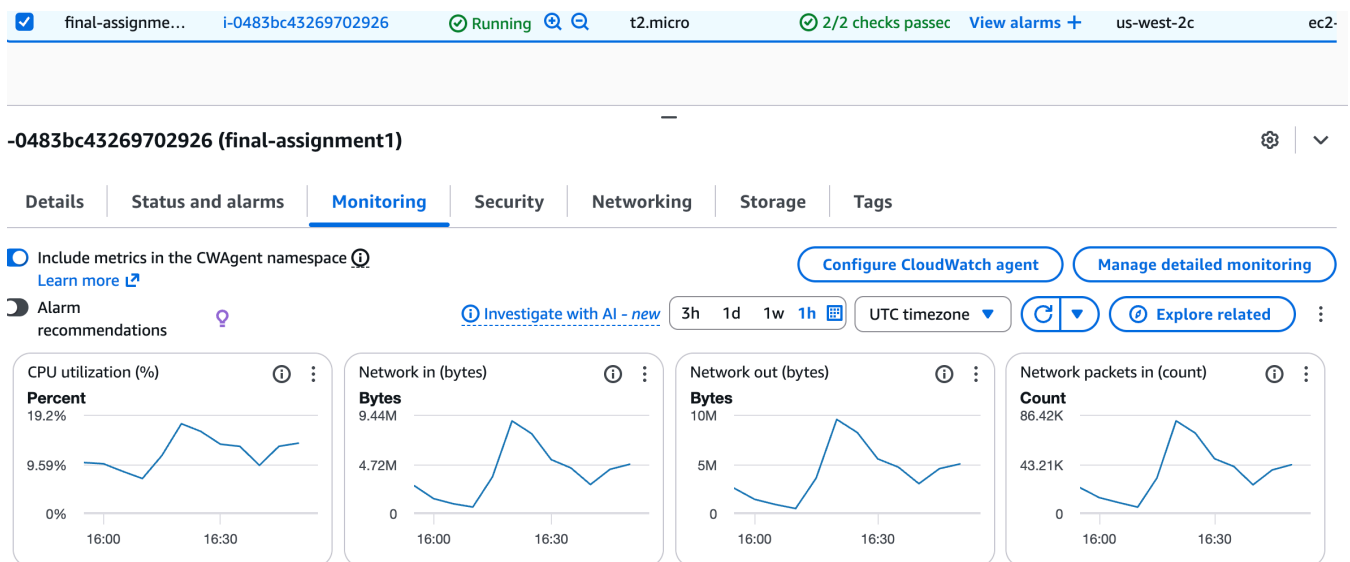
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Performance analysis charts



Evidence of EC2 deployment (EC2 console screenshot)

```
ubuntu@ip-172-31-3-1:~$ java -jar websocket-server-1.0-SNAPSHOT.jar 8080
Health check endpoint started on port 8081/health
ChatFlow Server starting on port 8080
ChatFlow WebSocket Server started successfully!
Listening on port: 8080
New connection to room: 1 from /107.122.84.129:9076
Connection closed for room: 1
```



i-0483bc43269702926 (final-assignment1)

⚙

- Details
- Status and alarms
- Monitoring
- Security
- Networking
- Storage
- Tags

▼ Instance summary

Info

Instance ID  i-0483bc43269702926	Public IPv4 address  100.23.97.14 open address ↗	Private IPv4 addresses  172.31.3.1
IPv6 address —	Instance state ✓ Running	Public DNS  ec2-100-23-97-14.us-west-2.compute.amazonaws.com open address ↗
Hostname type IP name: ip-172-31-3-1.us-west-2.compute.internal	Private IP DNS name (IPv4 only)  ip-172-31-3-1.us-west-2.compute.internal	